

KAMENSKOYE, Russian Federation

WMO: 257450

Lat: 62.467N

Lon: 166.213E

Elev: 132

StdP: 14.63

Time Zone: 12.00 (E12)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.5	-39.3	-47.1	0.4	-42.0	-43.7	0.4	-39.0	50.5	-1.2	45.3	0.9	2.8	100	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	72.7	58.7	69.5	57.5	66.2	56.1	60.7	69.2	58.9	66.8	57.3	63.9	8.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	69.5	63.2	55.4	65.6	60.4	54.1	62.5	59.4	27.0	69.4	25.8	66.7	24.7	64.1	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.4	23.7	20.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	-10.7	-7.2	-2.2	16.5	36.3	51.5	57.8	53.4	42.2	24.7	6.4	-6.8	
	DBStd	27.64	17.36	16.91	14.49	11.85	7.09	5.50	4.41	5.26	6.65	11.49	15.55	17.79	
	HDD50	10705	1882	1601	1620	1006	425	46	2	27	243	783	1308	1762	
	HDD65	15708	2347	2021	2085	1456	889	405	227	361	684	1248	1758	2227	
	CDD50	476	0	0	0	0	1	91	243	132	9	0	0	0	
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0	
	CDH74	70	0	0	0	0	0	10	48	12	0	0	0	0	
	CDH80	8	0	0	0	0	0	1	2	5	0	0	0	0	
Wind	WSAvg	8.4	9.8	10.3	9.2	8.4	8.2	8.4	7.5	7.2	6.4	7.5	9.1	8.7	
Precipitation	PrecAvg	16.2	1.4	1.2	0.9	0.8	0.7	1.2	1.9	2.2	1.7	1.2	1.3	1.2	
	PrecMax	28.5	4.7	7.0	2.9	2.3	1.7	3.8	5.0	3.9	5.6	3.1	6.0	4.0	
	PrecMin	9.8	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.4	0.0	0.3	0.2	0.0	
	PrecStd	4.3	1.1	1.2	0.7	0.6	0.4	0.8	1.2	0.9	1.1	0.6	0.9	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	32.4	36.9	41.5	61.3	74.6	78.7	72.8	62.7	46.8	38.2	37.3	
		MCWB	32.8	29.6	33.4	36.3	48.8	56.6	61.0	59.7	54.8	43.0	35.1	34.7	
	2%	DB	29.1	27.1	29.7	37.2	54.9	70.2	74.9	68.8	57.8	44.6	34.9	29.1	
		MCWB	26.6	25.8	26.9	33.3	44.4	55.5	59.7	58.4	50.8	41.3	32.8	27.5	
	5%	DB	20.1	23.0	25.0	35.3	50.7	66.6	71.4	65.1	54.8	41.2	31.7	23.4	
		MCWB	18.8	22.1	23.4	32.3	42.3	54.0	58.9	56.6	49.6	38.2	30.0	22.4	
	10%	DB	12.7	17.7	18.8	32.9	46.9	63.4	68.2	62.2	52.3	38.2	26.8	18.0	
		MCWB	11.7	17.1	17.3	30.3	40.3	52.4	57.8	55.5	48.1	35.6	25.3	17.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	30.0	33.2	37.1	48.9	58.2	63.1	62.9	55.2	43.8	35.9	34.5
			MCDB	36.0	31.8	36.2	41.5	60.7	70.8	74.5	69.9	60.2	45.9	37.3	37.3
2%		WB	26.3	25.8	27.0	34.3	45.1	56.3	61.3	60.1	52.7	41.3	33.2	27.8	
		MCDB	29.1	27.0	29.4	36.8	53.8	67.9	71.1	66.0	55.8	44.3	34.9	29.2	
5%		WB	18.8	22.1	23.3	32.4	42.6	54.9	59.9	58.0	50.5	38.6	29.7	22.5	
		MCDB	19.9	23.1	25.0	35.1	50.0	65.3	69.0	63.1	53.8	40.9	31.6	23.4	
10%		WB	11.7	17.0	17.5	30.4	40.6	53.1	58.4	56.3	48.8	35.9	25.4	17.2	
		MCDB	12.7	17.7	18.8	32.9	46.3	62.0	66.4	60.5	52.0	37.6	26.8	18.0	
Mean Daily Temperature Range	5% DB	MDBR	10.9	12.1	13.8	13.9	12.4	16.7	16.1	14.1	13.5	10.1	11.5	11.1	
		MCDBR	15.1	11.4	11.9	10.4	18.6	24.3	24.2	20.0	16.9	9.3	13.4	13.8	
		MCWBR	13.7	10.2	10.2	7.9	10.2	11.7	11.7	10.4	10.9	6.8	12.3	12.8	
	5% WB	MCDBR	15.2	11.4	12.0	10.1	17.6	21.5	21.1	15.2	13.3	7.9	12.9	13.7	
		MCWBR	13.8	10.4	10.3	7.7	9.8	11.1	10.9	8.8	9.1	6.1	11.9	12.7	
Clear-Sky Solar Irradiance	taub	0.197	0.257	0.317	0.384	0.452	0.414	0.498	0.476	0.337	0.293	0.222	0.181		
	taud	2.021	2.132	2.027	1.874	1.852	2.215	1.971	1.993	2.427	2.264	2.004	1.922		
	Ebn at Noon	156	221	242	246	239	256	226	217	243	204	132	91		
	Edh at Noon	10	21	35	50	56	40	50	45	24	18	10	6		
All-Sky Solar Radiation	RadAvg	88	355	908	1577	1633	1666	1580	1136	731	359	120	39		
	RadStd	12	37	70	75	190	188	154	127	71	49	18	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page